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UNSW SCIENCE FAIR 2026

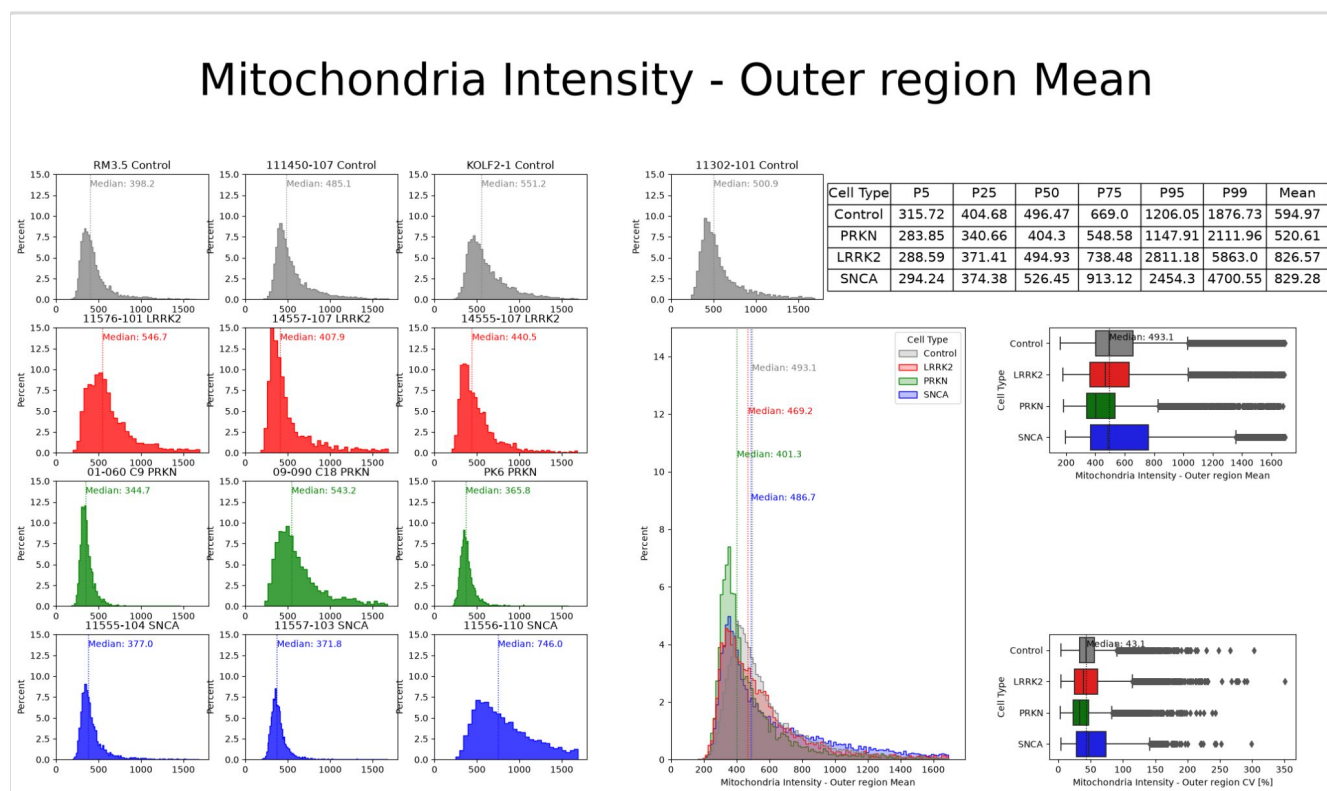
RUSHIK SRIRAMOJU

Gautam Wali team, Sue Lab, NeuRA

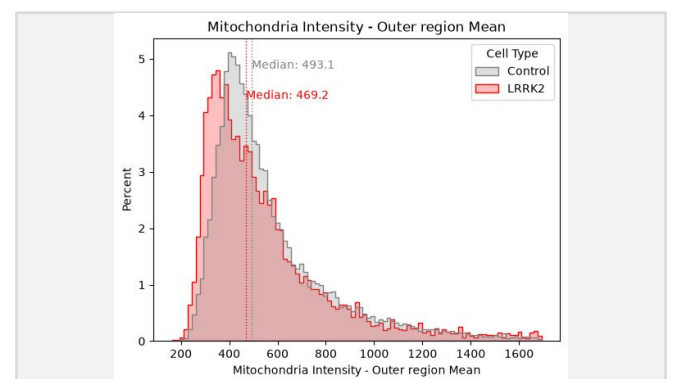
What am I doing?

Why do Parkinson's cells show altered mitochondrial function? I analyse individual cells to determine whether this change occurs across the whole cell population or is driven by a vulnerable subgroup of cells with particularly low mitochondrial membrane potential

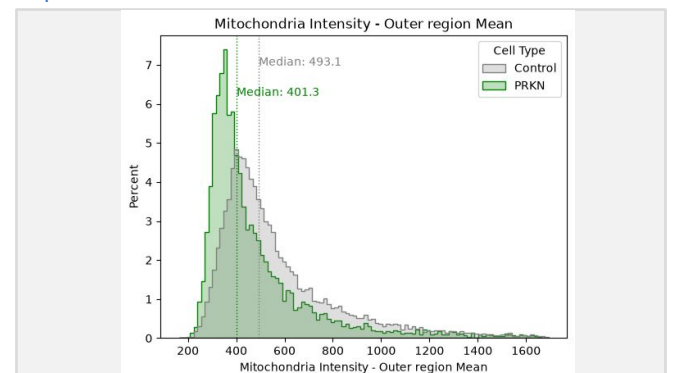
RESEARCH RESULTS



Single-cell analysis reveals different patterns of mitochondrial dysfunction across Parkinson's disease cell lines.



Population-wide shift observed with LRRK2



Increased percentage of cells with low TMRM in PRKN.

Why does this matter?

Average measurements can hide small but important groups of vulnerable cells. Identifying these subgroups gives us a more accurate picture of mitochondrial dysfunction and may help us determine whether a treatment improves the whole population or only certain cells.



My research interest

I study how single-cell analysis and machine learning can reveal hidden differences between cells and improve our understanding of mitochondrial dysfunction in Parkinson's disease.

3rd Year Bachelor of Science (Computer Science)/ Bachelor of Science (Bioinformatics)

NeuRA — Neuroscience Research Australia

UNSW Science Fair 2026 · Research group or laboratory

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QR
CODE